MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code: **MG1DSCHTC100** Course Title: **Basic horticulture**

Time: 1.5 Hrs **Maximum Marks: 50**

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part **A**:
Short answers
Answer ***any ten*** questions
Each question carries ***1 mark***

	Questions	Taxonomy	CO
1	What are the key differences between horticulture and agriculture?	Understand	1
2	What are the objectives of horticultural education and research in advancing the field?	Analyse	1
3	Wire a short note on Semi-arid fruit zone	Remember	2
4	Differentiate living and nonliving components of garden	Analyse	2
5	Which soil property is most important in determining irrigation frequency?	Analyse	3
6	What are the four R's in nutrient management?	Remember	3
7	What is the key principle of nutrient management?	Understand	3
8	What is a formal garden?	Remember	2
9	How can horticultural therapy be used in healthcare settings to improve physical and mental well-being	Skill	1
10	What principle governs the timing of irrigation in agriculture?	Remember	3
11	What is inarching?	Understand	5
12	What is serpentine layering?	Understand	5

(10×01=10)

Part **B**:
Short Essay Type Questions
Answer ***any six*** Questions
Each question carries ***5 marks***

13	Describe the process of creating a garden layout plan.	Understand	2
14	How do modern innovations like automated systems enhance the efficiency of greenhouses?	Evaluate	2
15	How evapotranspiration does influence the need for irrigation?	Apply	3
16	What is your understanding about Plant-Water-Soil	Evaluate	3

	relationship?		
17	How do modern technologies like genetic engineering and biotechnology influence plant breeding in horticulture?	Skill	1
18	Describe briefly about current trends and innovations in horticulture.	Evaluate	1
19	What are the applications of micro-propagation?	Apply	5
20	Compare the energy requirements of hydroponics and aquaponics.	Analyse	5

(6×5=30)

Part **C**:
Essay Type Questions
Answer ***any one*** question
Question carries ***10 marks***

21	Create a plan for implementing AI-driven precision agriculture in a small-scale farm	Create	5
22	Explain the role of plant growing structures like hotbeds, greenhouses, and rain shelters in sustainable agriculture, focusing on how they help mitigate environmental challenges.	Understand	2

(1×10=10)